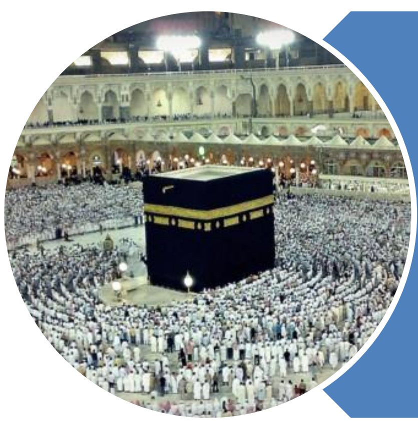
Data Structures - I

CPCS 204

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي اَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

STACKS

Introduction

Dr. Jonathan (Yahya) Cazalas

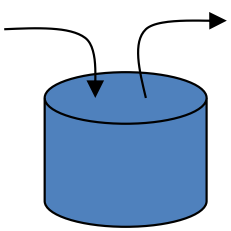
Dr. Muhammad Umair Ramzan

CPCS 204

Stacks - Introduction

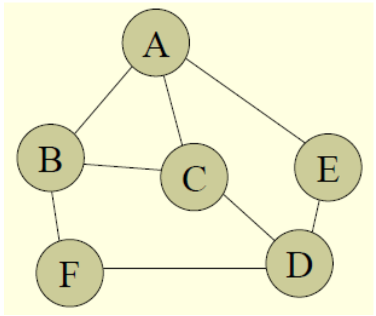
- Arrays

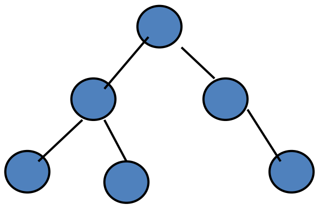
- Linked List



- Stacks

- Queues
- Tree
- Graph

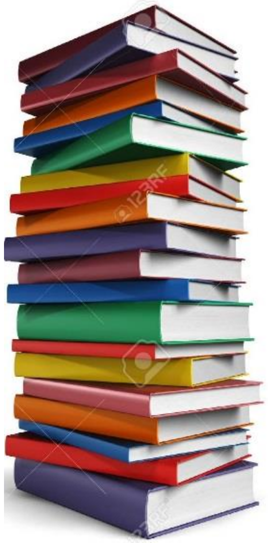




CPCS 204

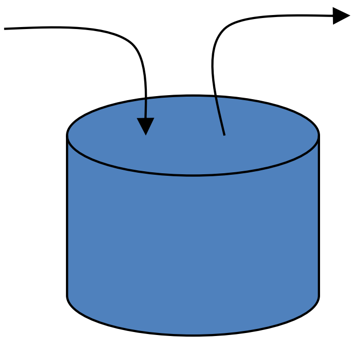
Stacks - Introduction

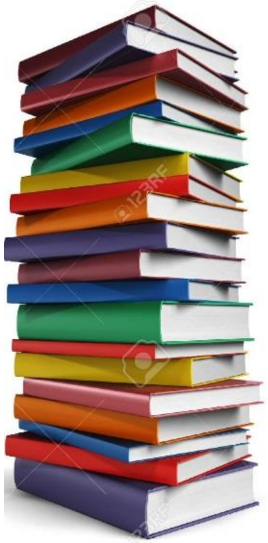
- Definition
  - A data structure that stores information in the form of a stack
  - Placing a value one over the other
- Example
  - Stack of trays in cafeteria
  - Stack of books
  - Stack of Servers



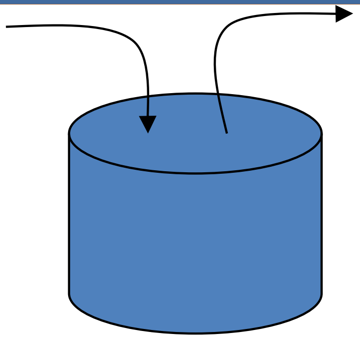
Stacks - Introduction

- Place a value at top
- Remove a value from top
- Abstract Data Type (ADT)
- Logical data structure
- Implementation
  - Arrays
  - Linked List



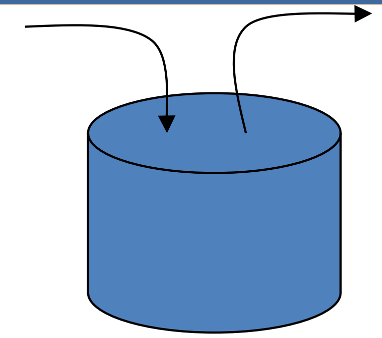


Stacks - Introduction

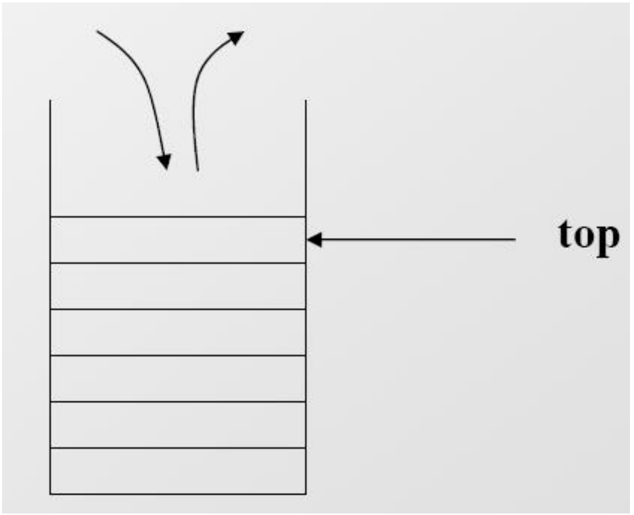


- Stacks are an Abstract Data Type
  - So for our discussion, we say they are not built into JAVA
- We must define them and their behaviors
- So what is a stack?
  - A data structure that stores information in the form of a stack.
  - Consists of a variable number of homogeneous elements
    - i.e. elements of the same type

Stacks - Introduction

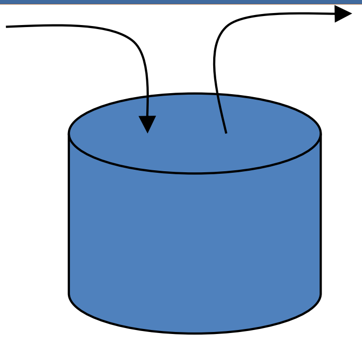


- **Access Policy**
  - the first element to be removed from the stack is the last element that was placed onto the stack
- **Last In First Out**
  - LIFO
- **Processing End**
  - TOP



CPCS 204

Stacks – Summary



- Abstract Data Type (ADT)
- Logical Data Structure
- Linear Data Structure
- Homogeneous
- Accessible only from one end i. e. Top
- Access policy is LIFO
- Implemented using Arrays or Linked List

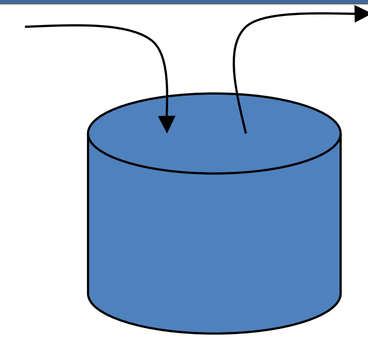
STACKS

Basic Operations
&
Its Use

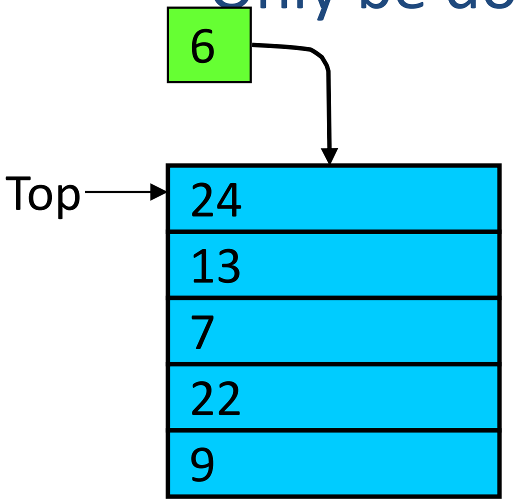
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

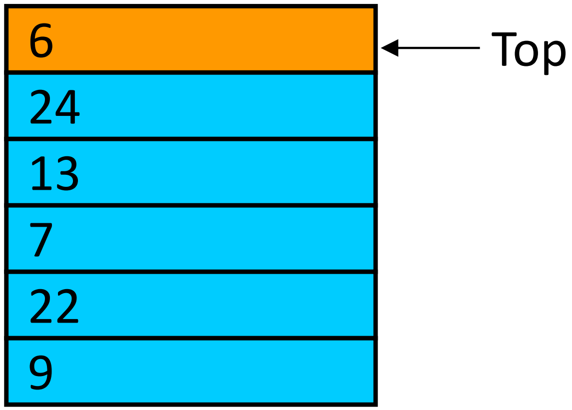
Stacks – Basic Operation



- **PUSH**
  - This pushes an item on top of the stack
    - Only be done if stack is not full

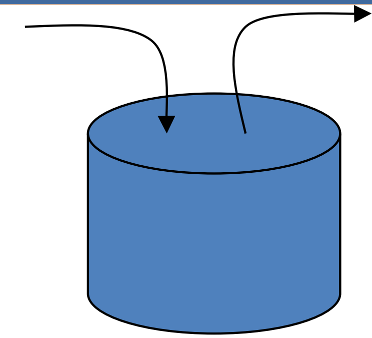


Stack S
(before push)

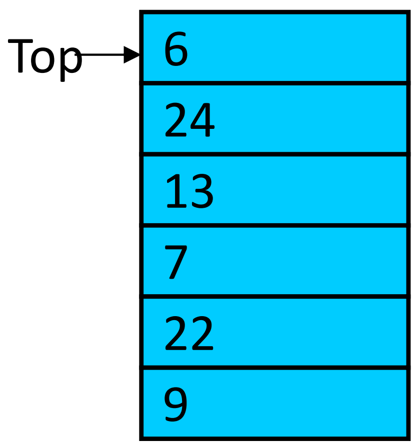


Stack S
(after push)

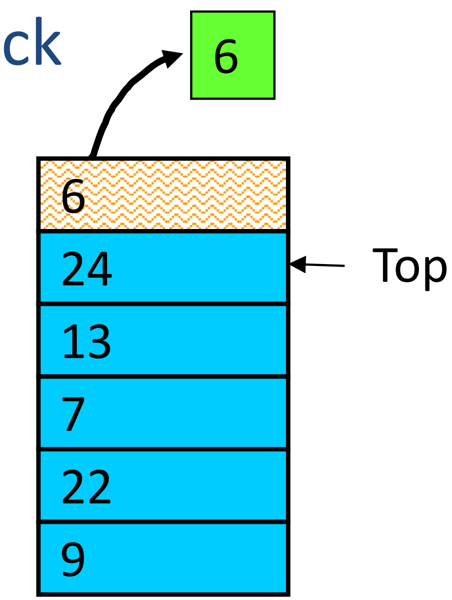
Stacks – Basic Operation



- **POP**
  - This pops an item from top of the stack
    - Only be done on a non-empty stack

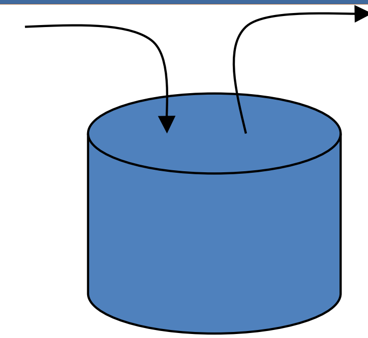


(before pop)

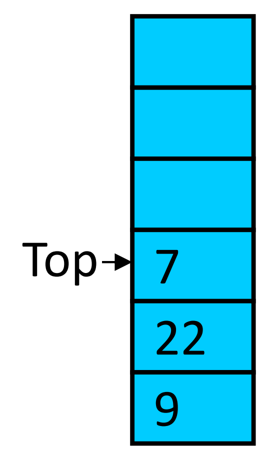


(after pop)

Stacks – Basic Operation

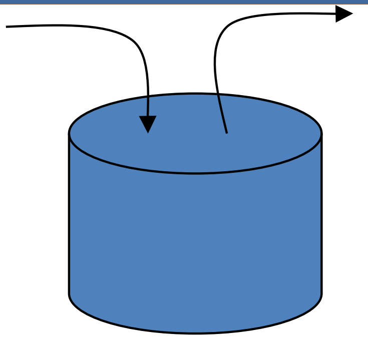


- **isEmpty**
  - Typically implemented as a **boolean** function
  - Returns **TRUE** if no items are in the stack
    - Top will be **-1** **(array)**
  - Returns **FALSE** if one or more items are in stack
    - Top will be greater or equal to **0** **(array)**
  - Needed for both **arrays** and **linked list**

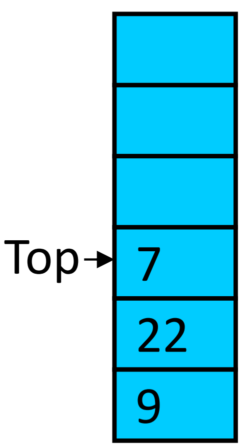


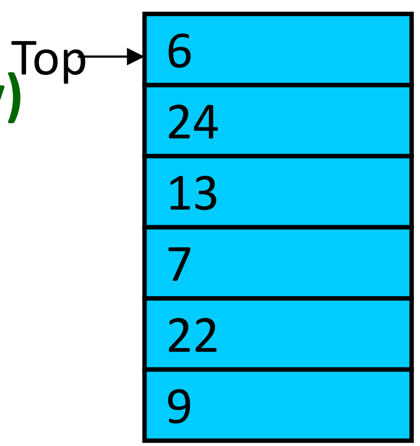
CPCS 204

Stacks – Basic Operation



- **isFull**
  - Typically implemented as a **boolean** function
  - Returns **TRUE** if stack has no further space
    - Top will be **length – 1 (array)**
  - Returns **FALSE** if one or more spaces are free in stack
    - Top will be less than **length – 1 (array)**
  - Needed only for **arrays**

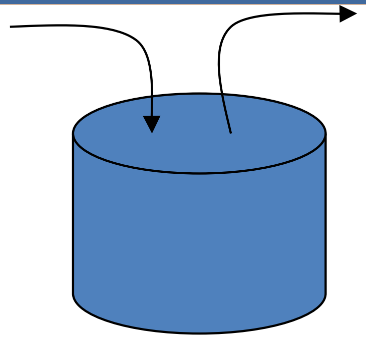




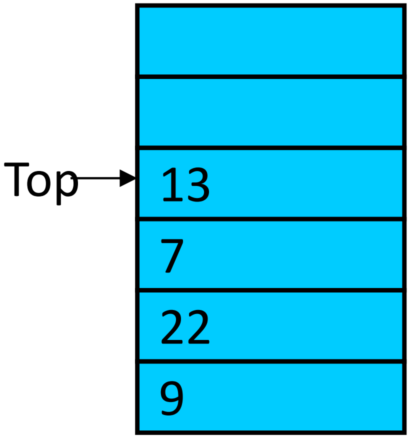
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

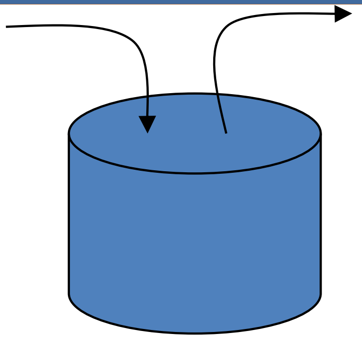
Stacks – Basic Operation



- **top**
  - Simply returns the **value** at the **top** of the stack without actually popping the stack
  - Value **13** will be returned

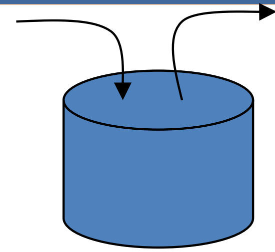


Stacks – Basic Operation



- **Push** Modify the content
- **Pop** Modify the content
- **isEmpty** Do not modify
- **isfull** Do not modify
- **top** Do not modify

Stacks – Usability



- When data needs to be stored and then retrieved in reverse order
- Example:
  - Implementation of RECURSION
  - Conversion of infix to postfix expression
  - Evaluation of postfix

Top - Down

Stack	Returned value
1	1
1*factorial(0)	1
2*factorial(1)	2
3*factorial(2)	6
4*factorial(3)	24
5*factorial(4)	120
factorial(5)	

Return 120

```
public static int factorial(int n) {
    if (n == 0)
        return 1;
    else
        return  n * factorial(n - 1);
}
```

RECURSIVE

5 * 3 + 2 + 6 * 4

5 3 * 2 + 6 4 * +

41

Dr. Jonathan (Yahya) Cazalas

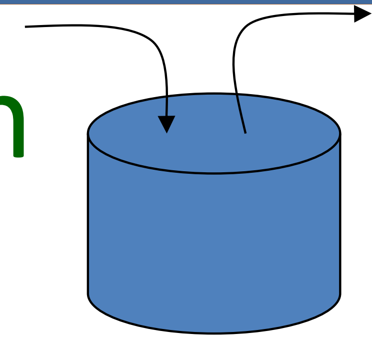
Dr. Muhammad Umair Ramzan

STACKS

Applications

CPCS 204

Applications - Introduction



- Evaluating Arithmetic Expressions

  5 * 3 + 2 + 6 * 4

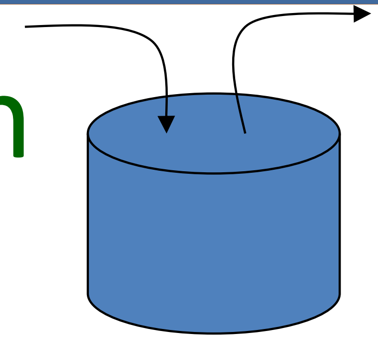
- Computer can solve the expression only Left to Right OR Right to Left

- Attempt – 1 (L-R)
  - 5 * 3 + 2 + 6 * 4
  - 15 + 2 + 6 * 4
  - 17 + 6 * 4
  - 23 * 4
  - **92**

- Attempt – 2 (R-L)
  - 5 * 3 + 2 + 6 * 4
  - 5 * 3 + 2 + 24
  - 5 * 3 + 26
  - 5 *29
  - **145**

CPCS 204

Applications - Introduction



- Evaluating Arithmetic Expressions

  5 * 3 + 2 + 6 * 4

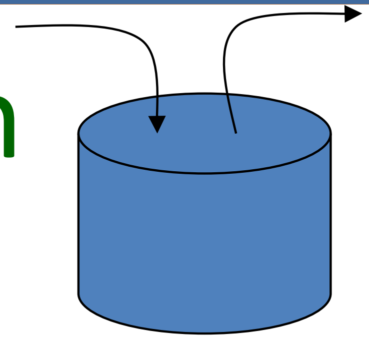
- Computer can solve the expression only Left to Right OR Right to Left

- Correct Solution
  - 5 * 3 + 2 + 6 * 4
  - 15 + 2 + 6 * 4
  - 15 + 2 + 24
  - 17 +24
  - **41**

- Example (Exercise)
  - 5 * (3 + (2 + 6))* 4

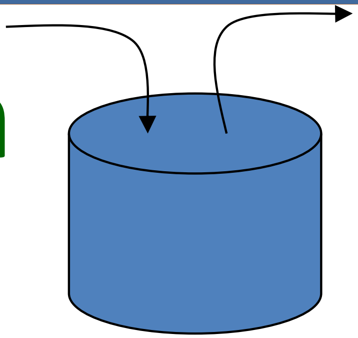
CPCS 204

Applications - Introduction



- Arithmetic Expression
  - Infix Notation
    - Operator comes between operands
    - A + B
  - Postfix Notation
    - Operator follows operands
    - A B +
  - Prefix Notation
    - Operator comes before operands
    - +A B

Applications - Introduction



- Infix Notation
  - Easy for humans to understand and evaluate
  - Apply precedence and associativity rules
  - Difficult to process for computers
- Postfix Notation
  - Easy for computer to understand and evaluate
  - precedence and associativity are embedded with the expression
  - No parenthesis in expression
  - Evaluated by scanning from Left to Right
  - Difficult to process for humans

CPCS 204

STACK Applications

Postfix Evaluation

Applications – Postfix Evaluation

- **Algorithm**
- Scan postfix expression completely symbol by symbol from left to right
  - If the symbol is "Operand"
    - Push to the stack
  - If symbol is "operator"
    - Pop first value from stack i.e pop1
    - Pop second value from stack i.e. pop2
    - Apply "operator" i.e pop2 ⊗ pop1
    - Push the result back to stack
- The value present at top of stack is the final result

Applications – Evaluating Postfix

- **Algorithm**
- Scan postfix expression symbol by symbol from left to right
  - If the symbol is "Operand"
    - Push to the stack
  - If symbol is "operator"
    - Pop first value from stack i.e pop1
    - Pop second value from stack i.e. pop2
    - Apply "operator" i.e pop2 ⊗ pop1
    - Push the result back to stack
- The value present at top of stack is the final result

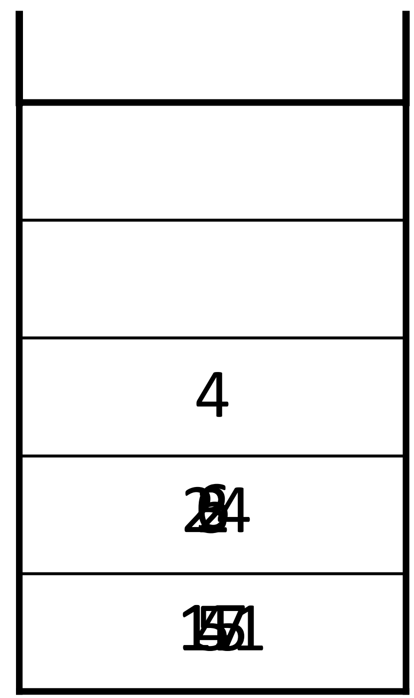
- 3 4 +
- 3 4 +
- 3 4 +
- 3 4 +
- $\boxed{3}$ + $\boxed{4}$
- 7

4
7

5 3 * 2 + 6 4 * +

- **Algorithm**
- Scan postfix expression symbol by symbol from left to right
  - If the symbol is "Operand"
    - Push to the stack
  - If symbol is "operator"
    - Pop first value from stack i.e pop1
    - Pop second value from stack i.e. pop2
    - Apply "operator" i.e pop2 ⊗ pop1
    - Push the result back to stack
- The value present at top of stack is the final result

41



Postfix Evaluation – Summary

- Expression will be scanned Left to Right
- Postfix expression contains
  - Operands i.e. A, K, 5, 8, etc
  - Operators i.e. +, *, /, -, etc
- Stack will be used to store
  - Operands
  - Results

CPCS 204

Postfix Evaluation

Examples

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

Calculation

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

Calculation

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

Calculation

7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

Calculation

7

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

8
7

Calculation

"No Bonus" marks for this course anymore

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

8
7

Calculation

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

Calculation

10
8
7

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

3
10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

3
10
8
7

Calculation

Data Structures - I

"No Bonus" marks for this course anymore

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

2
3
10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

2
3
10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 A 3 2 * - B / - 10 C + **Example 01**

10
8
7

A

Calculation
2

3
10
8
7

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation	
3	2

10
8
7

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

3 * 2 = 6

10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

3 * 2 = 6

6
10
8
7

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation

6
10
8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

Calculation
6

10
8
7

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

Calculation	
10	6

8
7

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

Calculation				
10	-	6	=	4

8
7

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

C
4
8
7

Calculation				
10	-	6	=	4

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

Calculation

4
8
7

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

Calculation
4

8
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

A
10
8
7

B
4
8
7

C
7

Calculation	
8	4

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

7

Calculation				
8	/	4	=	2

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

2
7

Calculation

8 / 4 = 2

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

4
8
7

B

2
7

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

A
10
8
7

B
4
8
7

Calculation
2

C
7

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

10
8
7

A

4
8
7

B

C

Calculation

7 - 2 = 5

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

A
10
8
7

B
4
8
7

Calculation				
7	-	2	=	5

5

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

C
5

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

C P C S 2 0 4

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

C
10
5

10
5

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + **Example 01**

A B C

A
10
8
7

B
4
8
7

C
10
5

10
5

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A	B	C
10	4	
8	8	10
7	7	5

5

Calculation
10

7 8 10 3 2 * - / - 10 + Example 01

A B C

A
10
8
7

B
4
8
7

C
10
5

Calculation	
5	10

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A	B	C
10	4	
8	8	10
7	7	5

Calculation

5 + 10 = 15

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

10
8
7

A

4
8
7

B

10
5

C

Calculation				
5	+	10	=	15

15

CPCS 204

7 8 10 3 2 * - / - 10 + Example 01

A B C

A	B	C
10	4	
8	8	10
7	7	5

A B C

Calculation

15

7 8 10 3 2 * - / - 10 + Example 01

A B C

A	B	C
10	4	
8	8	10
7	7	5

Result

15

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

3 8 7 - 6 * +

A B

Example 02

Calculation

3 8 7 - 6 * +

A B

Example 02

3

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

3 8 7 - 6 * +

A B

Example 02

3

Calculation

3 8 7 - 6 * +

A B

Example 02

8
3

Calculation

3 8 7 - 6 * +

A B

Example 02

8
3

Calculation

CPCS 204

3 8 7 - 6 * +

A B

Example 02

7
8
3

Calculation

3 8 7 _ - 6 * +

A B

Example 02

7
8
3

A

Calculation

7
8
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation

7
8
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation
7

8
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation

8 7

3

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation
8 - 7 = 1

3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation				
8	-	7	=	1

1
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation

1
3

3 8 7 - 6 * +

A B

Example 02

7
8
3

A

Calculation

6
1
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 02

3 8 7 - 6 * +

A B

7
8
3

A

Calculation

6
1
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

Calculation
6

B
1
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

A

Calculation

1 6

3

C P C S 2 0 4

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

A

Calculation				
1	*	6	=	6

B
3

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

B
6
3

Calculation

1 * 6 = 6

3 8 7 - 6 * _ +

A B

Example 02

A
7
8
3

B
6
3

6
3

Calculation

CPCS 204

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

A

B
6
3

B

6
3

Calculation

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

B
6
3

3

Calculation
6

Example 02

3 8 7 - 6 * +

A B

A
7
8
3

B
6
3

Calculation	
3	6

CPCS 204

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

B
6
3

Calculation

3 + 6 = 9

CPCS 204

3 8 7 - 6 * +

A B

Example 02

A
7
8
3

B
6
3

9

Calculation

3 + 6 = 9

CPCS 204

3 8 7 - 6 * + _

A B

Example 02

A
7
8
3

B
6
3

A B

Calculation

9

CPCS 204

3 8 7 - 6 * + _

A B

Example 02

A
7
8
3

B
6
3

Result
9

"No Bonus" marks for this course anymore

CPCS 204

Example 03

					1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

1 above 7, 2 above 8, 3 above −

6 12 2 / 3 7 + 2 8 5 + * − + +

Calculations

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

Calculations

6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Calculations

6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Data Structures - I

"No Bonus" marks for this course anymore

Example 03

				1				2				3				
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+		

12
6

Calculations

CPCS 204

Example 03

					1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

Stack
12
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

						1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	

Calculations

2
12
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Calculations

2
12
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

					1				2			3			
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+	

Example 03

12
6

Calculations
2

CPCS 204

Example 03

			1				2				3			
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Calculations	
12	2

6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

6

Calculations

12 / 2 = 6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Calculations

12 / 2 = 6

6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

				1					2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	

Example 03

6
6

Calculations

CPCS 204

				1					2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	+

Example 03

3
6
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

					1							3			
								2							
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	

Example 03

3
6
6

1

Calculations

3
6
6

CPCS 204

Example 03

					1							2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

3
6
6

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

3
6
6

1

Calculations

7
3
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

CPCS 204

Example 03

						1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	

3
6
6

1

Calculations

7
3
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

				1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Example 03

3
6
6

1

Calculations

7

3
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

3 7

6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

3 + 7 = 10

6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

3 + 7 = 10

10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

								1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Example 03

3
6
6

1

Calculations

10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

						1		2			3			
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

2
10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

3
6
6

1

Calculations

2
10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

	1				2			3						
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

8
2
10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

				1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

3
6
6

1

Calculations

8
2
10
6
6

Example 03

			1				2					3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1
3
6
6

2
8
2
10
6
6

Calculations

8
2
10
6
6

C P C S 2 0 4

Example 03

			1					2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

Stack 1:

3
6
6

1

Stack 2:

8
2
10
6
6

2

Calculations

8
2
10
6
6

CPCS 204

Example 03

				1					2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+	

1	2
	8
	2
3	10
6	6
6	6

5
8
2
10
6
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

C P C S 2 0 4

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

1
3
6
6

2
8
2
10
6
6

5
8
2
10
6
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

CPCS 204

Example 03

$$6 \quad 12 \quad 2 \quad / \quad \overset{1}{3} \quad 7 \quad + \quad \overset{2}{2} \quad 8 \quad 5 \quad \underline{+} \quad * \quad \overset{3}{-} \quad + \quad +$$

1	2
	8
	2
3	10
6	6
6	6

Calculations

5

8
2
10
6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1				2				3	
6	12	2	/	3	7	+	2	8	5	+	*	-	+	+

1	2
	8
	2
3	10
6	6
6	6

2
10
6
6

Calculations

8 5

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					[1]				[2]			[3]		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2
	8
	2
3	10
6	6
6	6

2
10
6
6

Calculations

8 + 5 = 13

CPCS 204

Example 03

				1				2			3			
6	12	2	/	3	7	+	2	8	5	**+**	*	–	+	+

Stack 1

3
6
6

Stack 2

8
2
10
6
6

Calculations

8 + 5 = 13

Stack 3

13
2
10
6
6

Example 03

				1					2			3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1
3
6
6

2
8
2
10
6
6

13
2
10
6
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

1 2 3

6 12 2 / 3 7 + 2 8 5 + * − + +

3
6
6

1

8
2
10
6
6

2

2
10
6
6

Calculations
13

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2
	8
	2
3	10
6	6
6	6

10
6
6

Calculations

2 13

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*****	−	+	+

1	2
	8
	2
3	10
6	6
6	6

10
6
6

Calculations

2 * 13 = 26

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2
	8
	2
3	10
6	6
6	6

26
10
6
6

Calculations

2 * 13 = 26

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

$6 \quad 12 \quad 2 \quad / \quad 3 \quad \overset{1}{7} \quad + \quad 2 \quad 8 \quad \overset{2}{5} \quad + \quad * \quad \overset{3}{-} \quad + \quad +$

1	2
	8
	2
3	10
6	6
6	6

26
10
6
6

Calculations

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

	8
	2
3	10
6	6
6	6
1	2

10
6
6

Calculations

26

Example 03

6 12 2 / 3 7 + 2 8 5 + * − + +

(superscripts above the expression: 1 over "7", 2 over "5", 3 over "−")

1	2
	8
	2
3	10
6	6
6	6

6
6

Calculations

10 26

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

1	2
	8
	2
3	10
6	6
6	6

6
6

Calculations

10 - 26 = -16

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

				^{1}			^{2}				^{3}		
6	12	2	/	3 7	+	2	8 5	+	*	–	+	+	

1	2
	8
	2
3	10
6	6
6	6

Calculations

10 - 26 = -16

-16
6
6

CPCS 204

Example 03

					1			2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

1	2
	8
	2
3	10
6	6
6	6

-16
6
6

Calculations

10 - 26 = -16

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	–	+	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

-16
6
6

CPCS 204

Example 03

$6 \quad 12 \quad 2 \quad / \quad 3 \quad \overset{1}{7} \quad + \quad 2 \quad 8 \quad \overset{2}{5} \quad + \quad * \quad - \quad \overset{3}{\underline{+}} \quad +$

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

-16
6
6

Calculations

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

6	12	2	/	3	7	+	2	8	5	+	*	–	+	+
				1				2				3		

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations
-16

6
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

				1				2					3	
6	12	2	/	3	7	+	2	8	5	+	*	–	<u>+</u>	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

6 (-16)

6

Example 03

1 2 3

6 12 2 / 3 7 + 2 8 5 + * − + +

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

6

Calculations

6 + (-16) = -10

Example 03

6	12	2	/	3	7 (1)	+	2	8	5 (2)	+	*	−	+ (3)	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

6 + (-16) = -10

-10
6

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Example 03

					1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

-10
6

Calculations

Example 03

1 2 3

6 12 2 / 3 7 + 2 8 5 + * – + +

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

(-10)

6

CPCS 204

Example 03

				1				2					3	
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

6 (-10)

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

CPCS 204

Example 03

					1				2			3		
6	12	2	/	3	7	+	2	8	5	+	*	−	+	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

6 + (-10) = -4

CPCS 204

Example 03

				1				2				3		
6	12	2	/	3	7	+	2	8	5	+	*	-	+	+

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

-4

Calculations

6 + (-10) = -4

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Example 03

6 12 2 / 3 7 + 2 8 5 + * − + + _

(operator markers above the expression: 1 over 7, 2 over 8, 3 over −)

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Calculations

-4

Example 03

6 12 2 / 3 $\overset{1}{7}$ + 2 8 $\overset{2}{5}$ + * $\overset{3}{-}$ + + _

1	2	3
	8	
	2	
3	10	-16
6	6	6
6	6	6

Result
- 4

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

C
P
C
S

2
0
4

STACK Applications

Infix to Postfix

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Introduction

- Infix expression may contains
  - Operands i.e. A, K, 5, 8, etc
  - Operators i.e. +, *, /, -, etc
  - Opening Parenthesis i.e. (
  - Closing Parenthesis i.e.)
- Postfix will contains only
  - Operands
  - Operators

5 * (3 + (2 + 6))* 4

5 3 2 6 + + * 4 *

CPCS 204

Infix to Postfix – Rules

Symbol	Action
	Scan the expression from left to right
Operand A,K,6 etc	Add to postfix expression e.g. **A**
Operator +, *, - etc	
Empty	**1)** Incoming operator is pushed to stack e.g. **+**
Non - Empty	**2)** If incoming operator has **greater precedence** as compared to operator on the top then **push** it to the stack e.g. *****
	3) If incoming operator has **equal precedence** as compared to operator on the top then **pop** it and **add to postfix** expression e.g. **/**; then go to **step 1** if stack is empty otherwise to **step 2**
	4) If incoming operator has **less precedence** as compared to operator on the top then **pop it** and **add to postfix** expression e.g **.-** ; then go to **step 1** if stack is empty otherwise to **step 2**

~~*~~ /
+

Postfix Expression
A * / +

CPCS 204

Infix to Postfix – Rules

Symbol	Action
(	**Outside the stack** its precedence is highest, so **push** to stack ***Inside stack** its precedence is **lowest**
)	**Pop** the operators **one by one** and **add to postfix** expression until **(** found.
	Discard both of parenthesis
	If expression is **scanned completely**, **pop all values one by one** and **add to postfix** expression

Stack
(
-

Stack
/
*
-
(
-

Postfix Expression
A C / K + G A / * - -

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

5*3+2+6*4

Symbol	Action
	Scan the expression from left to right
Operand A,K,6 etc	Add to postfix expression e.g. A
Operator +, *, - etc	
Empty	1) Incoming operator is pushed to stack e.g. +
Non - Empty	2) If incoming operator has greater precedence as compared to operator on the top then push it to the stack e.g. *
	3) If incoming operator has equal precedence as compared to operator on the top then pop it and add to postfix expression e.g. /; then go to step 1 if stack empty otherwise to step 2
	4) If incoming operator has less precedence as compared to operator on the top then pop it and add to postfix expression e.g .- ; then go to step 1 if stack empty otherwise to step 2
(	Outside the stack its precedence is highest, so push to stack
)	Pop the operators one by one and add to postfix expression until (found.
	Discard both of parenthesis
	If expression is scanned completely, pop all values one by one and add to postfix expression

Postfix Expression

5 3 * 2 + 6 4 * +

STACK
~~*~~
~~+~~
~~+~~
~~*~~

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Summary

- Expression will be scanned Left to Right
- Infix expression may contains
  - Operands i.e. A, K, 5, 8, etc
  - Operators i.e. +, *, /, -, etc
  - Opening Parenthesis i.e. (
  - Closing Parenthesis i.e.)
- Postfix will contains only
  - Operands
  - Operators
- Stack will be used only for
  - Operators

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix

Examples

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

Postfix Expression

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

Postfix Expression

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

Postfix Expression
M

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – **Example 01**

M + (P – K) * T

A B C

Postfix Expression
M

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression
M

+

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression

M

+

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression
M

(
+

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression
M

(
+

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression
M P

(
+

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

Postfix Expression
M P

(
+

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

-
(
+

B

-
(
+

Postfix Expression
M P

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

-
(
+

B

-
(
+

Postfix Expression
M P

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

A
+

B
-
(
+

-
(
+

A

B

Postfix Expression
M P K

Infix to Postfix – Example 01

M + (P – K) * T

A B C

A
+

B
-
(
+

Postfix Expression
M P K

-
(
+

CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

A
- +

B
- -
- (
- +

C
- (
- +

Postfix Expression

M P K -

Infix to Postfix – Example 01

M + (P – K) * T

A B C

+

A

-
(
+

B

+

Postfix Expression

M P K -

Infix to Postfix – Example 01

M + (P – K) * T

A B C

Stack A
+

A

Stack B
-
(
+

B

Postfix Expression
M P K -

+

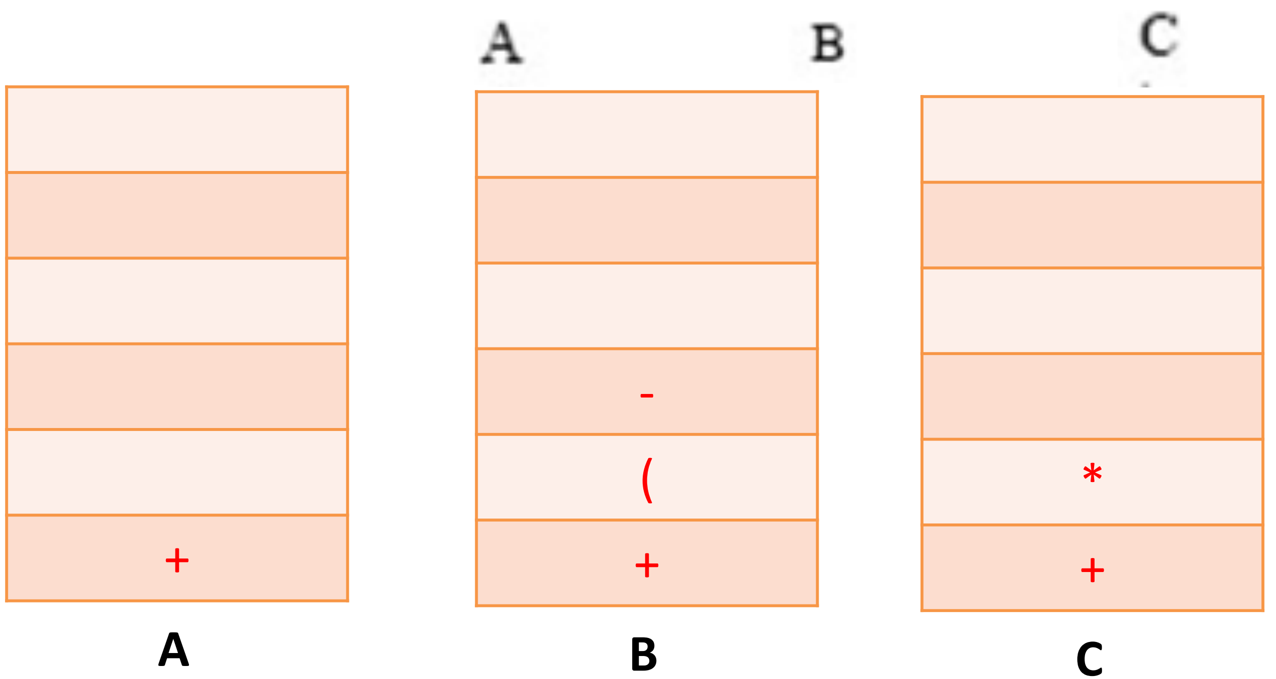
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

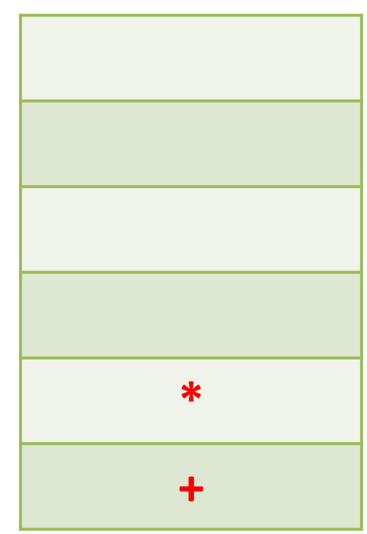
Infix to Postfix – Example 01

M + (P – K) * T



Postfix Expression

M P K -



CPCS 204

Infix to Postfix – Example 01

M + (P – K) * T

A B C

-
(
+

*
+

A

+

B

-
(
+

C

*
+

Postfix Expression

M P K -

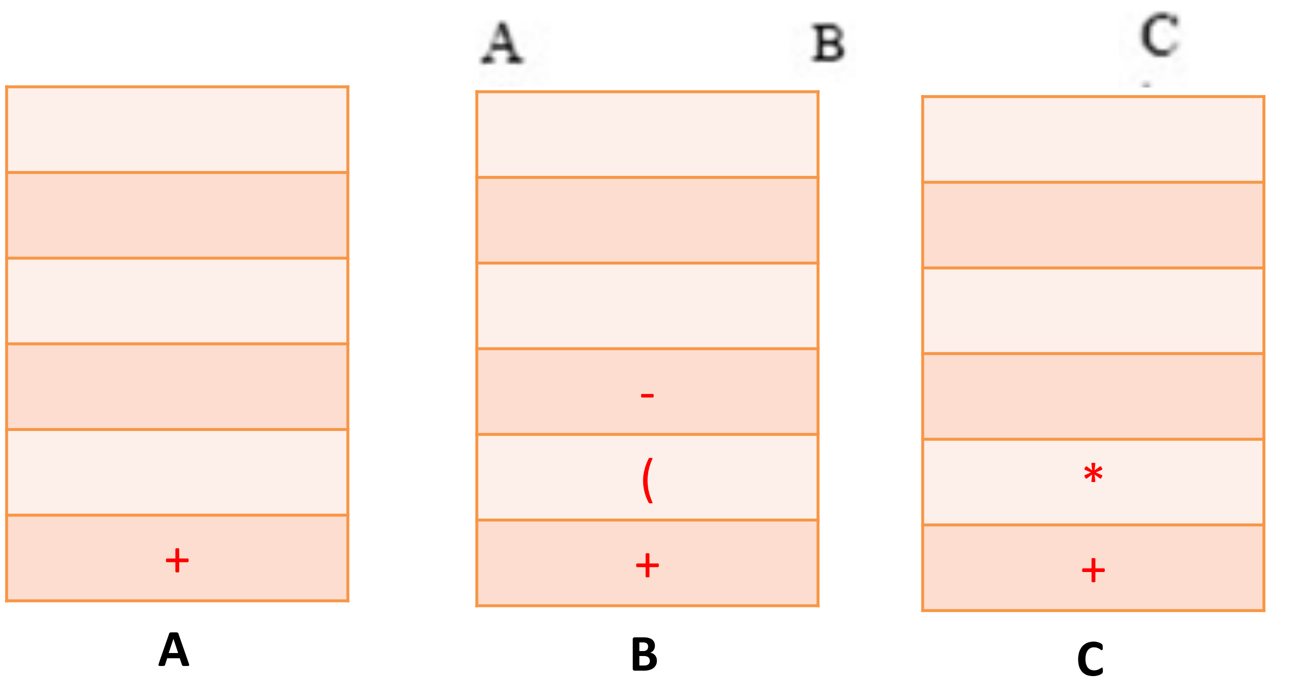
*
+

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

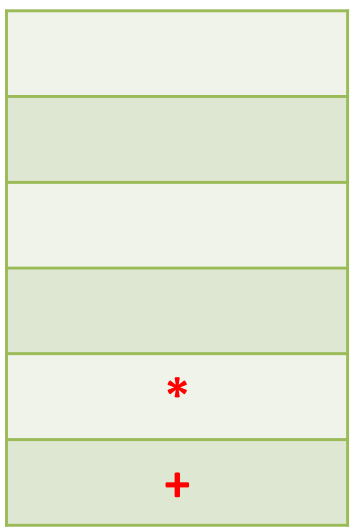
Infix to Postfix – Example 01

M + (P – K) * T



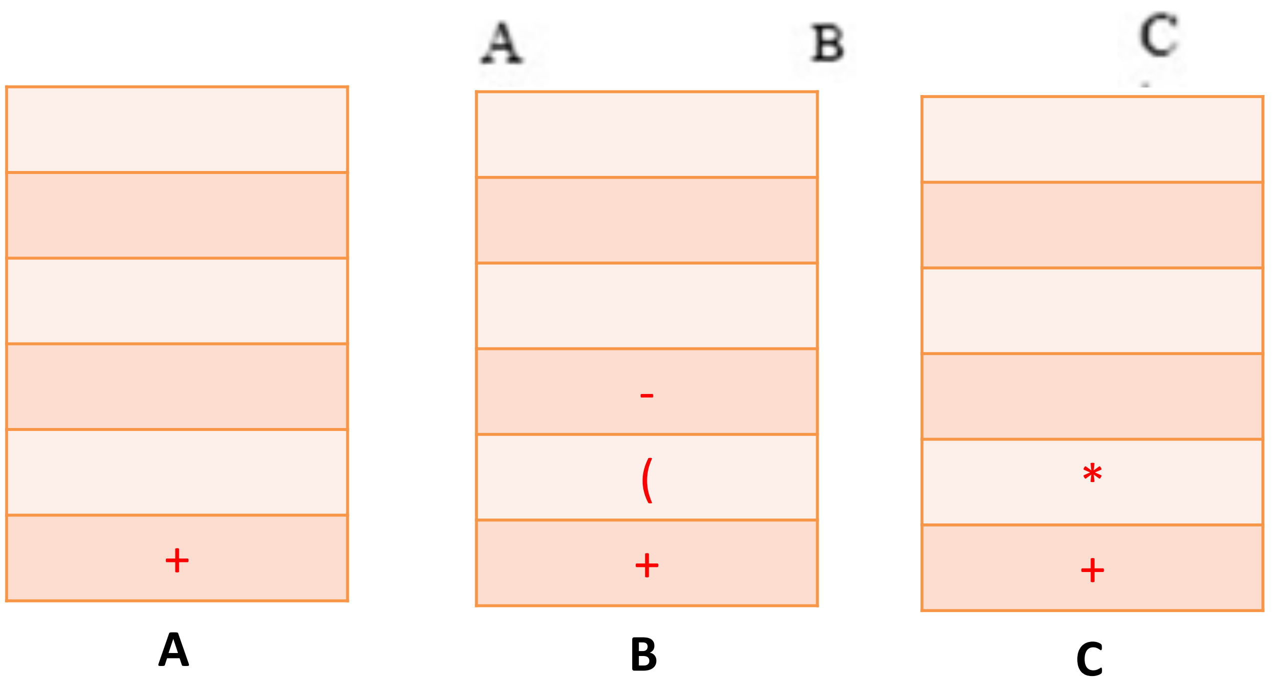
Postfix Expression

M P K - T



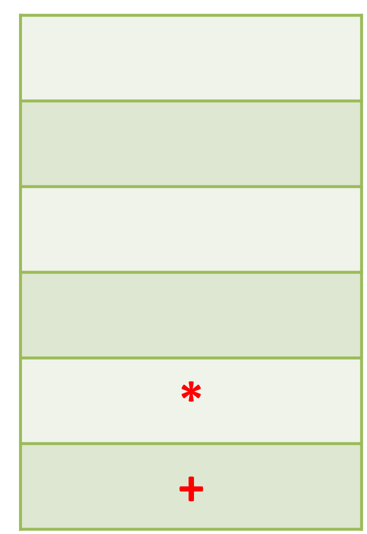
Infix to Postfix – Example 01

M + (P – K) * T



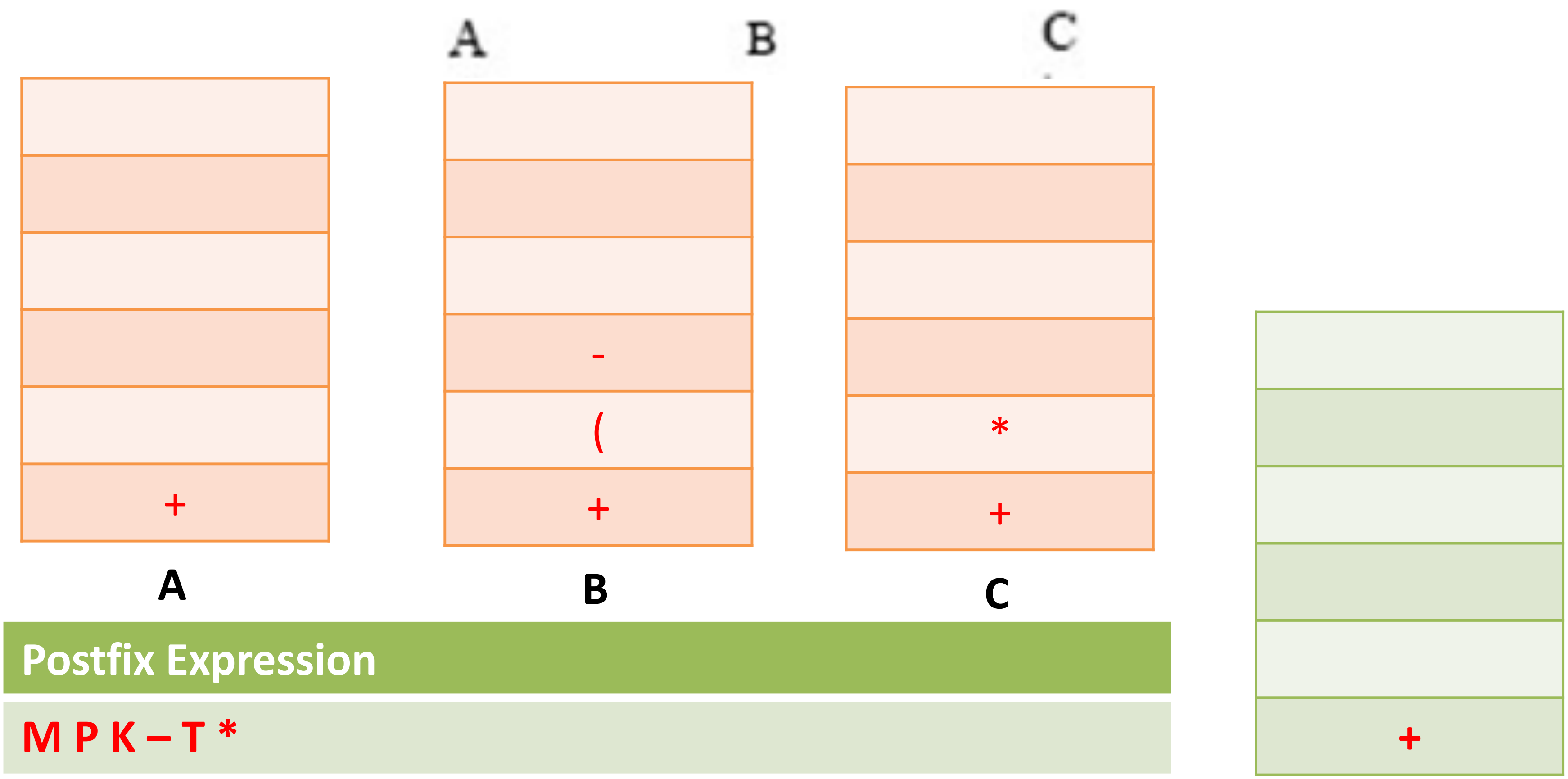
Postfix Expression

M P K - T



Infix to Postfix – Example 01

M + (P – K) * T



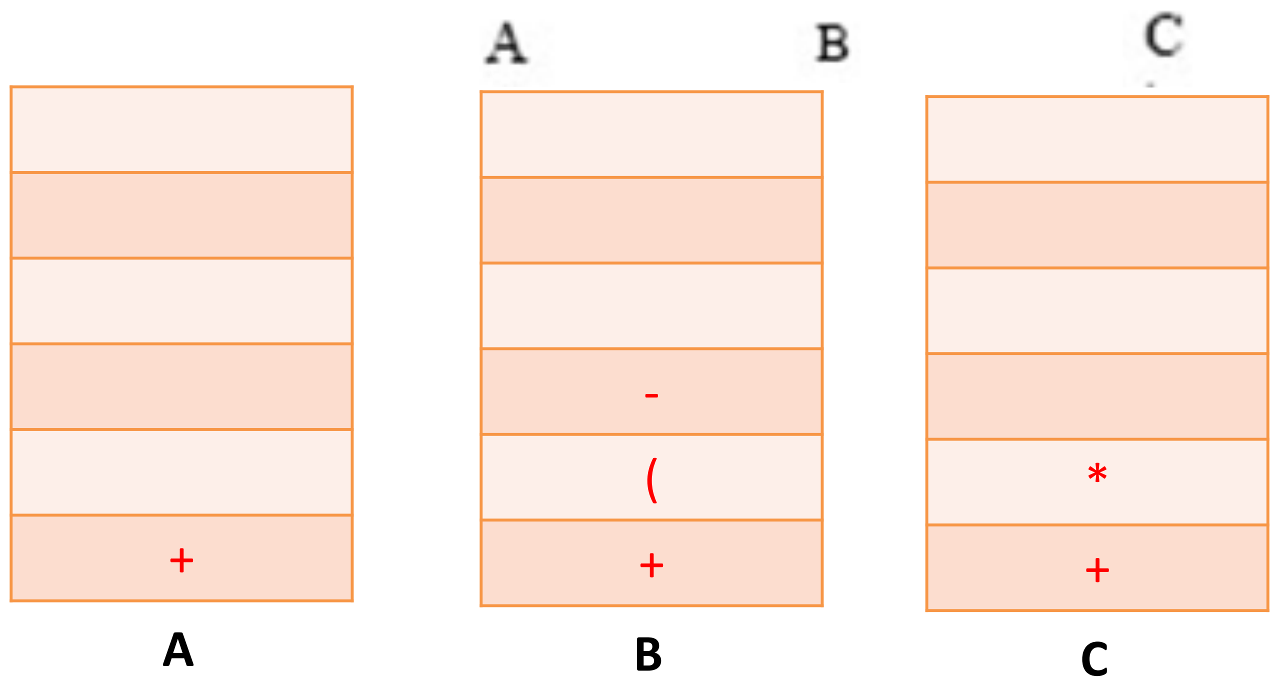
Postfix Expression

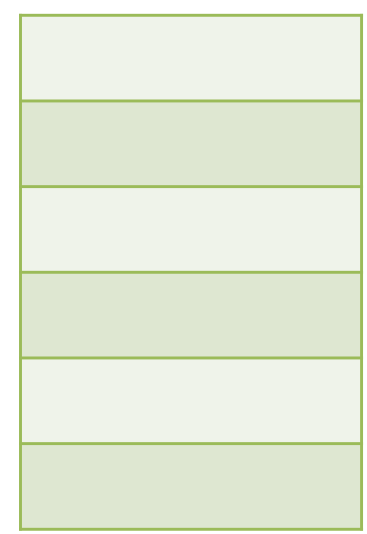
M P K – T *

CPCS 204

Infix to Postfix – **Example 01**

M + (P – K) * T





Postfix Expression
M P K – T * +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M- P/(T-K *N)+ R

A B C

Postfix Expression

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

Postfix Expression

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

Postfix Expression
M

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

Postfix Expression
M

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M – P / (T – K * N) + R

A B C

Postfix Expression
M

Stack
-

CPCS 204

Infix to Postfix – Example 02

M – P / (T – K * N) + R

A B C

Postfix Expression
M

Stack
-

CPCS 204

Infix to Postfix – Example 02

M – P / (T – K * N) + R

A B C

-

Postfix Expression
M P

CPCS 204

Infix to Postfix – Example 02

M− P/(T−K *N)+ R

A B C

Postfix Expression
M P

-

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K*N)+ R

A B C

Postfix Expression
M P

Stack
/
-

CPCS 204

Infix to Postfix – Example 02

M – P / (T – K * N) + R

A B C

Postfix Expression
M P

/
-

C P C S 2 0 4

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P

(
/
-

CPCS 204

Infix to Postfix – Example 02

M – P / (T – K * N) + R

A B C

(
/
-

A

Postfix Expression
M P

(
/
-

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T

(
/
-

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T

(
/
-

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T

-
(
/
-

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

-
(
/
-

Postfix Expression

M P T

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T K

-
(
/
-

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T K

-
(
/
-

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T K

*
-
(
/
-

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M− P/(T−K *N)+ R

A B C

(
/
-

A

Postfix Expression
M P T K

*
-
(
/
-

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

*
-
(
/
-

Postfix Expression

M P T K N

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

"No Bonus" marks for this course anymore

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

*
-
(
/
-

Postfix Expression

M P T K N

Dr. Jonathan (Yahya) Cazalas

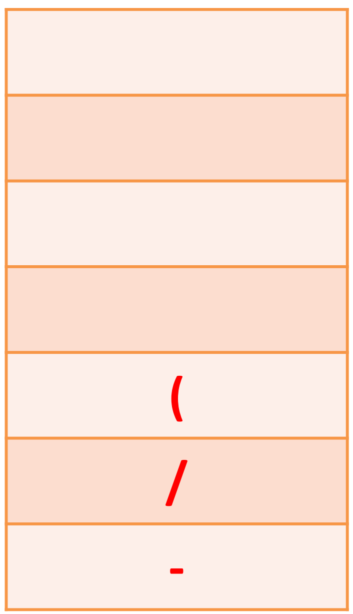
Dr. Muhammad Umair Ramzan

CPCS 204

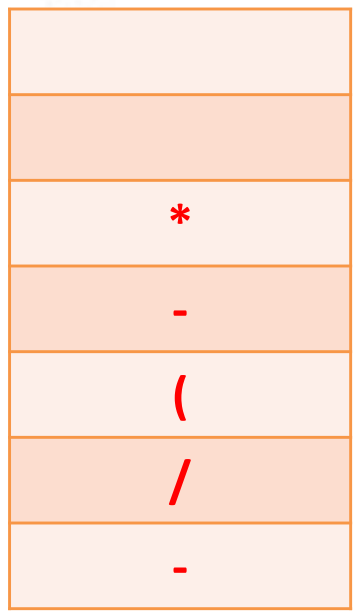
Infix to Postfix – Example 02

M− P/(T−K *N)+ R

A B C

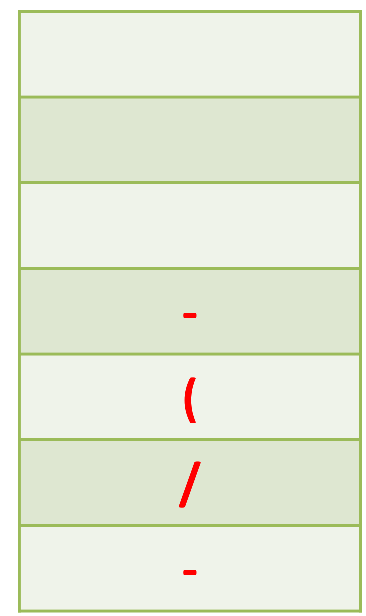


A



B

Postfix Expression
M P T K N *



Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 02

M- P/(T-K *N)+ R

A B C

(
/
-

A

*
-
(
/
-

B

(
/
-

Postfix Expression

M P T K N * -

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

Postfix Expression

M P T K N * -

/
-

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

/
-

Postfix Expression

M P T K N * -

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A	B
	*
	-
(	(
/	/
-	-

Postfix Expression
M P T K N * - /

-

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

Postfix Expression

M P T K N * - / -

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A	B	C
	*	
	-	
(	(	
/	/	
-	-	+

+

Postfix Expression

M P T K N * - / -

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 02

M– P/(T–K *N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

C
+

+

Postfix Expression

M P T K N * - / -

CPCS 204

Infix to Postfix – **Example 02**

M- P/(T-K*N)+ R

A B C

A
(
/
-

B
*
-
(
/
-

C
+

+

Postfix Expression
M P T K N * - / - R

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 02

M- P/(T-K *N)+ R

A B C

A	B	C
	*	
	-	
(	(	
/	/	
-	-	+

+

Postfix Expression

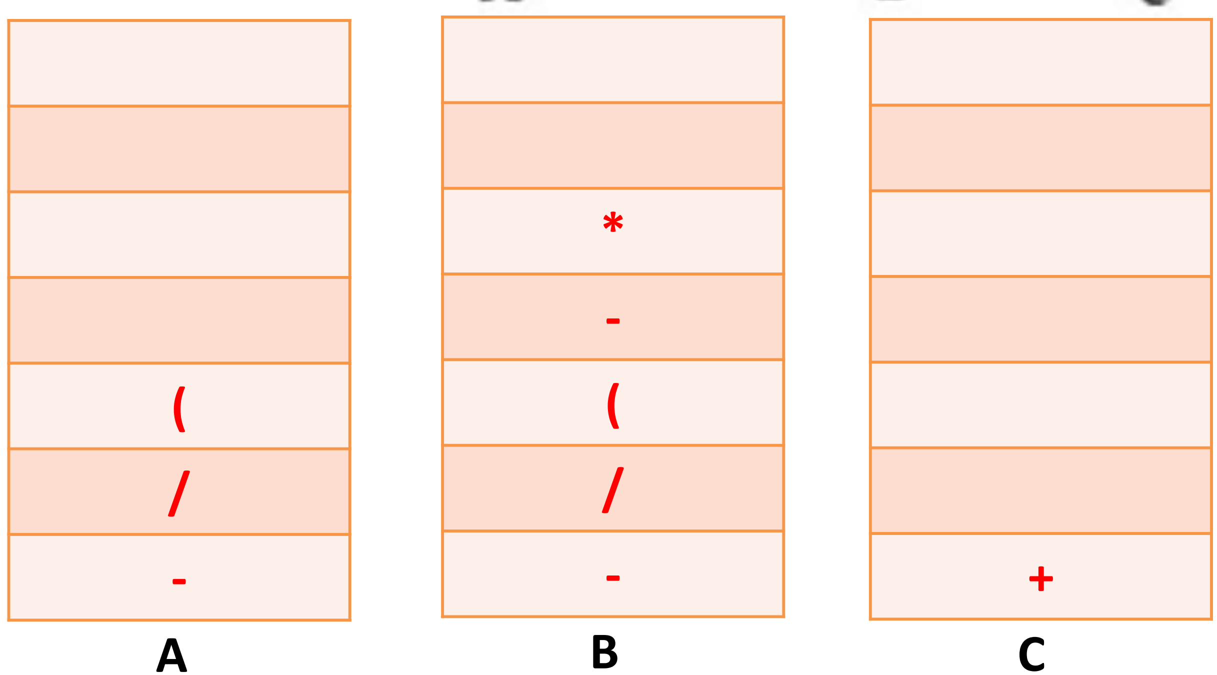
M P T K N * - / - R

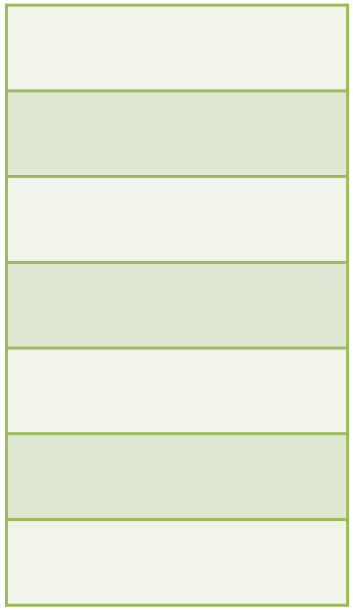
CPCS 204

Infix to Postfix – Example 02

M- P/(T-K *N)+ R

A B C





Postfix Expression

M P T K N * - / - R +

Infix to Postfix – Example 03

```
                  1                           2                                   3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Postfix Expression

Infix to Postfix – Example 03

```
                    1                         2                                 3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Postfix Expression

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                    2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Postfix Expression

Stack
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                        1                 2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Postfix Expression

Stack
(

CPCS 204

Infix to Postfix – Example 03

```
                  1                        2                          3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Postfix Expression
A

Stack
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                    2                  3
 (  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Postfix Expression
A

Stack
(

"No Bonus" marks for this course anymore

CPCS 204

Infix to Postfix – Example 03

```
                        1                 2                    3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Postfix Expression
A

Stack
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                        2                          3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Stack
+
(

Postfix Expression
A

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                        1                                  2                                            3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack
(
+
(

Postfix Expression
A

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                      1                   2                        3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Postfix Expression
A

Stack
(
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                        2                           3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Postfix Expression
A B

Stack
(
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                                  2                                    3
(   A   +   (   B   *   C   )   +   D   )   /   (   E   +   F   )   *   G   -   H   /   I
```

Postfix Expression
A B

Stack
(
+
(

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                  1                         2                            3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

Postfix Expression
A B

*
(
+
(

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                  1                     2                         3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression
A B

*
(
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                1                    2                          3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

Postfix Expression
A B C

*
(
+
(

Infix to Postfix – Example 03

```
                  1                    2                            3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression
A B C

*
(
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – **Example 03**

```
                 1                       2                          3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression

A B C *

(
+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – **Example 03**

```
              1                   2                            3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*

(

+

(

1

Postfix Expression

A B C *

+

(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                 1                    2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

Postfix Expression

A B C *

+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                          1                            2                                    3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

(

Postfix Expression

A B C * +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                        1                             2                                    3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

+
(

Postfix Expression

A B C * +

Dr. Jonathan (Yahya) Cazalas
Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                  1                     2                           3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

+
(

Postfix Expression

A B C * +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                      1                              2                                      3
(   A   +   (   B   *   C   )   +   D   )   /   (   E   +   F   )   *   G   -   H   /   I
```

*
(
+
(

1

+
(

Postfix Expression

A B C * + D

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                            2                                 3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

Postfix Expression
A B C * + D

+
(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
            1                 2                     3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

Postfix Expression

A B C * + D +

(

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                      1                            2                                     3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression
A B C * + D +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                     1                            2                           3
 (  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression
A B C * + D +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                  1                          2                                    3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

Postfix Expression
A B C * + D +

/

CPCS 204

Infix to Postfix – Example 03

```
                         1                                2                                            3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

*
(
+
(

1

Postfix Expression
A B C * + D +

/

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
               1                        2                                 3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

Stack 1 (top to bottom): *, (, +, (

1

Stack 2 (top to bottom): (, /

2

Stack (top to bottom): (, /

Postfix Expression

A B C * + D +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                1                          2                          3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

2
(
/

Stack		
(		
/		

Postfix Expression

A B C * + D +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                                                           2                                       3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

2
(
/

(
/

Postfix Expression

A B C * + D + E

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                  1                          2                               3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1	2	
*		
(		
+	(	
(	/	
		(
		/

Postfix Expression

A B C * + D + E

Infix to Postfix – Example 03

```
                  1                    2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

2
(
/

+
(
/

Postfix Expression

A B C * + D + E

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

1 2 3

(A + (B * C) + D) / (E + F) * G – H / I

1
*
(
+
(

2
(
/

+
(
/

Postfix Expression

A B C * + D + E

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                     1                    2                          3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1	2
*	
(	
+	(
(	/

+
(
/

Postfix Expression

A B C * + D + E F

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                          1                                 2                                            3
 (   A   +   (   B   *   C   )   +   D   )   /   (   E   +   F   )   *   G   -   H   /   I
```

1	2
*	
(	
+	(
(	/

3
+
(
/

Postfix Expression
A B C * + D + E F

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                  1                               2                                        3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1
*
(
+
(

1

Stack 2
(
/

2

Stack 3
(
/

Postfix Expression

A B C * + D + E F +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                      1                             2                                    3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

*
(
+
(

1

(
/

2

/

Postfix Expression

A B C * + D + E F +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                      1                             2                           3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1
*
(
+
(

2
(
/

/

Postfix Expression

A B C * + D + E F +

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                1                    2                           3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1
*
(
+
(

1

Stack 2
(
/

2

Postfix Expression

A B C * + D + E F + /

CPCS 204

Infix to Postfix – Example 03

```
                1                   2                       3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

2
(
/

*

Postfix Expression

A B C * + D + E F + /

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                         2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1
*
(
+
(

1

Stack 2
(
/

2

*

Postfix Expression

A B C * + D + E F + /

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                1                          2                                3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1
*
(
+
(

2
(
/

*

Postfix Expression
A B C * + D + E F + / G

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                    1                            2                            3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1
*
(
+
(

2
(
/

*

Postfix Expression

A B C * + D + E F + / G

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                    1                           2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1
*
(
+
(

1

Stack 2
(
/

2

Postfix Expression

A B C * + D + E F + / G *

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                1                            2                                         3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  –  H  /  I
```

*
(
+
(

1

(
/

2

-

Postfix Expression

A B C * + D + E F + / G *

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                          1                  2                         3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1	2	
*		
(		
+	(	
(	/	-

Postfix Expression
A B C * + D + E F + / G *

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                       1                            2                                   3
 (   A   +   (   B   *   C   )   +   D   )   /   (   E   +   F   )   *   G   -   H   /   I
```

*
(
+
(

1

(
/

2

-

Postfix Expression

A B C * + D + E F + / G * H

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                1                         2                           3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1
*
(
+
(

1

Stack 2
(
/

2

Stack
-

Postfix Expression

A B C * + D + E F + / G * H

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Infix to Postfix – Example 03

```
                1                   2                          3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1	2	3
*		
(		
+	(	/
(	/	-

/
-

Postfix Expression

A B C * + D + E F + / G * H

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                         1                           2                         3
(  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1	2	3
*		
(		
+	(	/
(	/	-

/
-

Postfix Expression

A B C * + D + E F + / G * H

Infix to Postfix – Example 03

```
                     1                         2                           3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

Stack 1	Stack 2	Stack 3
*		
(		
+	(	/
(	/	-

Stack
/
-

Postfix Expression

A B C * + D + E F + / G * H I

Infix to Postfix – Example 03

```
                        1                    2                   3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1	2	3
*		
(		
+	(	/
(	/	-

/
-

Postfix Expression

A B C * + D + E F + / G * H I

CPCS 204

Infix to Postfix – Example 03

```
                     1                            2                                  3
( A + ( B * C ) + D ) / ( E + F ) * G - H / I
```

1	2	3
*		
(		
+	(	/
(	/	-

-

Postfix Expression

A B C * + D + E F + / G * H I /

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Infix to Postfix – Example 03

```
                  1                  2                             3
 (  A  +  (  B  *  C  )  +  D  )  /  (  E  +  F  )  *  G  -  H  /  I
```

1	2	3
(+ (*	/ (	- /

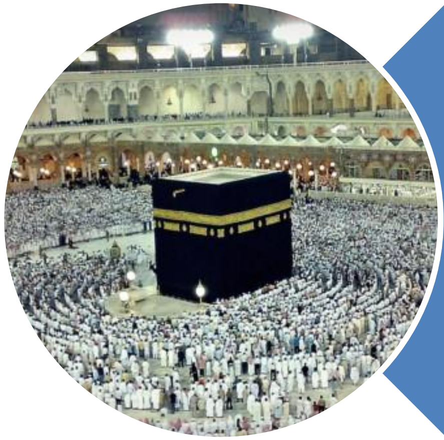
Postfix Expression

A B C * + D + E F + / G * H I / -

Data Structures - I

"No Bonus" marks for this course anymore

CPCS 204



Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us). **(Quran 3 : 173)**

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan